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Analysis of traffic accident size for Korean highway using structural equation models

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ABSTRACT

Accident size can be expressed as the number of involved vehicles, the number of damaged vehicles, the number of deaths and/or the number of injured. Accident size is the one of the important indices to measure the level of safety of transportation facilities. Factors such as road geometric condition, driver characteristic and vehicle type may be related to traffic accident size. However, all these factors interact in complicate ways so that the interrelationships among the variables are not easily identified. A structural equation model is adopted to capture the complex relationships among variables because the model can handle complex relationships among endogenous and exogenous variables simultaneously and furthermore it can include latent variables in the model. In this study, we use 2649 accident data occurred on highways in Korea and estimate relationship among exogenous factors and traffic accident size. The model suggests that road factors, driver factors and environment factors are strongly related to the accident size.
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1. Introduction

Traffic accident forecasting models have been developed to understand factors affecting traffic accidents and eventually to reduce traffic accidents by controlling and/or improve factors. In Korea, the total length of highways is over 3000 km and it is within the top 10 in the world. However, the number of accidents-per-one kilometer of highways is higher than any other countries. The rapid increase of travel demand and transportation infrastructures since 1980s, may have influenced on the high rates of traffic accident. Hence, it is a very interesting issue to identify reasons and/or factors making the trend of accidents in Korea.

Accident statistics most often used are to quantify and describe three principal informational elements: *accident occurrence*, *accident involvements* and *accident severity*. Accident occurrence relates to the numbers and types of accidents, accident involvements concerns the numbers and types of vehicles and drivers involved in accidents, and accident severity is generally expressed as the numbers of deaths and/or injuries occurring (William and Roger, 1990). While each statistic provides a meaningful information, an integrate information of accidents is also very useful. For instance, while vehicle factors such as the number of involved vehicles and the number of damaged vehicles in accidents can describe some aspects of accidents, unexplained aspects of accidents still exist. Hence, a new statistic "accident size" is adopted in this study, which can be described in terms of the number of deaths and injured persons as well as the number of damaged vehicles and the number of vehicles involved in accidents.

From the previous researches, factors such as road geometric conditions, driver characteristics and vehicle types can be related to accidents. However, all those factors interact in complicate way so that the interrelationships among the variables are not easily identified. A structural equations model (SEM) is adopted to capture the complex relationships among variables because the model can handle complex relationship among endogenous and exogenous variables simultaneously and furthermore, it can include latent variables in the models. In this study, we use 2649 accident data occurred on highways of Korea and estimate relationships among exogenous factors and traffic accident size. In modeling process, we create exogenous latent variables such as "road factors", "driver factors" and "environment factors" to identify latent relationships to an endogenous variable "accident size". This paper consists of six sections. Second section summarizes the previous works related to accident analysis and SEMs. Statistical methodologies adopted in this study are introduced in section three. Forth section describes the data in use, which is highway accident dataset of Korea. Several empirical models and interpretation of the results are shown in section five. Finally, conclusions and recommendation for future studies are drawn in section six.

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2. Previous works

Extensive effort has been made by many researchers in various transportation fields to explain traffic accident occurrence and factors affecting accidents. They attempted to develop different types of model, which can explain severity of accidents and eventually understand and predict accidents. Several selected studies related to ours are summarized among numerous studies, which are accident analysis and SEM. The description of SEM will be presented in next section.

For Hong Kong, the effects of factors influencing on the severity of injury from an accident was examined for factors such as human, vehicle, safety, environment and site. Risk factors associated with each of the vehicle types were identified by means of stepwise logistic regression models. For private vehicles, district board, gender of driver, age of vehicle, time of the accident and street light conditions were significant factors determining the severity of injury (Kelvin and Yau, 2004). Milton et al. (2008) proposed a mixed logit model using highway-injury data from Washington State. Findings in the study indicated that volume-related variables such as average daily traffic per lane, average daily truck traffic percentage, number of interchanges per mile and weather effects such as snowfall are best modelled as random-parameters, while roadway characteristics such as the number of horizontal curves, number of grade breaks per mile and pavement friction are best modelled as fixed parameters. Kim et al. (2007) conducted research for the factors contributing to the injury severity of bicyclists in bicycle–motor vehicle accidents using a multinomial logit model. The model predicted the probability of four injury severity outcomes: fatal, incapacitation, non-incapacitation, and possible or no injury. The results showed several factors, which more than double the probability of bicyclist suffering a fatal injury in an accident, all other things being kept constant. Notably, inclement weather, darkness with no streetlights, a.m. peak (06:00 a.m. to 09:59 a.m.), head-on collision, speeding-involved, vehicle speeds above 48.3 km/h, truck involved, intoxicated driver, bicyclist age 55 or over, and intoxicated bicyclist.

Many researches applying the SEMs can be found in transportation fields. These researches try to understand the complex relationships among the variable using SEM. Hamdar et al. (2008) developed a quantitative intersection aggressiveness propensity index (API). The index was intended to capture the overall propensity for aggressive driving to be experienced at a given signalized intersection. The index was a latent quantity that could be estimated from observed environmental, situational and driving behavior variables using SEM techniques. The exogenous variables were number of heavy vehicles, number of pedestrians, traffic volume, average queue length, percent grade, number of lanes, number of left turn lanes and so forth. Choo (2007) analyzed telecommunications impacts on travel in a comprehensive system considering demand, supply, costs, and land use, using SEM. The model results suggested that as telecommunications demand increases, travel demand increases, and vice versa. Additionally, transportation infrastructure and land use significantly affect travel demands.

In addition, SEM is frequently adopted in travel value and behavior field. Chung and Ahn (2002) developed SEM that presented relationships among socio-demographics, activity participation (i.e., time use), and travel behavior for each day during a week in a developing country. It was tentatively concluded that there were similar relationships between socio-demographics and travel behaviors in developing and developed countries. It was also confirmed that activity patterns were significantly different on weekdays and weekends. Furthermore, during weekdays there were day-to-day variations in the patterns of activity participation and travel behavior. Choi and Chung (2003) adopted multivariate SEM to handle the hierarchical nature of the data and explain complex relationship among socioeconomic factors of individuals and household, activity participation, and travel behavior using Puget Sound Transportation Panel data. Chung and Lee (2002) constructed an SEM to estimate aggregated automobile demand with data from Korea. The results indicated that both the number of driver's license holders and total road length had a statistically significant effect on automobile demands. In addition, several other determinants of the endogenous variables were found such as average household size, economically active population, personal transportation expenditure, urbanized area, and population density. Lu and Pas (1999) described the development, estimation and interpretation of a model relating socio-demographics, activity participation (time use) and travel behavior. Activity participation (time allocated to a number of activity types) and travel behavior were endogenous to the model. They reported the relationships between in-home and out-of-home activity participation and travel behavior.

3. Statistical methodology

SEM is a technique that can handle a large number of endogenous and exogenous observed variables simultaneously. Since SEM consists of a set of equations that are specified by direct links between variables, it can be called "the simultaneous equations" from the perspective. However, in SEM, we can introduce 'latent variables' which are the unobserved variables and represent unidimensional concepts in their purest form. Other terms for these are unobserved or unmeasured variables and factors. The observed variables of a latent variable contain random or systematic measurement errors, but the latent variable is free of these. Since all latent variables corresponding to concepts, they are hypothetical variables. Latent variables specified as linear combinations of the observed variables. The linear combinations are weighted averages. Hence, regression, path analysis, factor analysis and canonical correlation analysis are all special cases of SEM. In SEM we can separate errors in measurement from errors in equations (Golob, 2003).

3.1. Elements of SEM

A SEM with latent variables has at most three components as shown in Fig. 1: (a) a measurement model for the endogenous variables (Y measurement model), (b) a measurement model for the exogenous variable (X measurement model), and (c) a structural model.

The structural parameters are the elements of the three matrices. B is the matrix ($m \times m$) of direct effects among endoge-

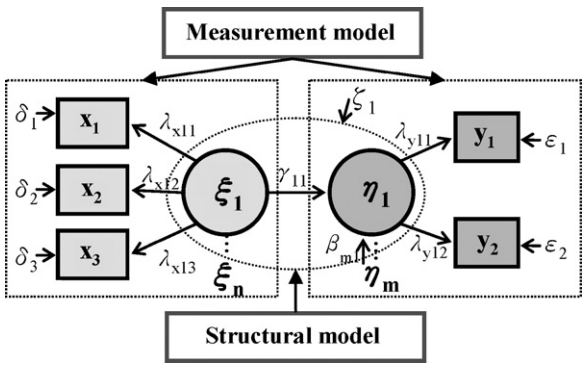


Fig. 1. An example of structural equation model.

Table 1
Elements of structural equation model

Measurement model	x	$q \times 1$ column vector of observed exogenous variables
	y	$p \times 1$ column vector of observed endogenous variables
	ξ	$n \times 1$ column vector of latent exogenous variables
	η	m $\times$ 1 column vector of latent endogenous variables
	δ	$q \times 1$ column vector of measurement error terms for observed variables x
	ε	$p \times 1$ column vector of measurement error terms for observed variables y
	Λ_X	The matrix $(q \times n)$ of structural coefficients for latent exogenous variables to their observed indicator variables
	Λ_Y	The matrix $(p \times m)$ of structural coefficients for latent endogenous variables to their observed indicator variables
Structural model	Γ	The matrix $(m \times n)$ of regression effects for exogenous latent variables to endogenous latent variables
	B	The coefficient matrix $(m \times m)$ of direct effects between endogenous latent variables
	ζ	$m \times 1$ column vector of the error terms
Covariance matrix	Θ_ε	The covariance matrix $(p \times p)$ of ε
	Θ_δ	The covariance matrix $(q \times q)$ of δ
	Φ	The covariance matrix $(n \times n)$ of ξ
	Ψ	The covariance matrix $(m \times m)$ of ζ

The β coefficients (components of B matrix) and the γ coefficients (components of Γ matrix) are magnitudes of expected changes after a unit increases in η or ξ. Similarly, λ coefficients (components of Λ matrix) are expected changes of observed variables with respect to a unit change in the latent variable.

nous latent variables and Γ is the matrix $(m \times n)$ of regression effects for exogenous latent variables to endogenous latent variables. Λ is linking matrix between the latent and observed variables. The elements of SEM are explained in Table 1.

3.2. Estimation methods

A SEM is applied in this research to estimate a simultaneous model that presents the interrelationships among latent variables: road factors, driver factors, environment factors and accidents size, which are explained by observed variables. The LISREL Version 8.51 and PRELIS/SIMPLIS software are used to estimate the model in this research. To estimate parameters in SEM, LISREL offers seven different methods: instrumental variables (IV), two-stage least squares (TSLS), unweighted least squares (ULS), generalized least squares (GLS), maximum likelihood (ML), weighted least squares (WLS), and diagonally weighted least squares (DWLS). Generally, ML method is most widely used as estimator of parameters because $(N-1)F_{ML}$ is approximately distributed in large samples (N) as Chi-square with an assumption of multivariate normality of variables. However, when distributions of variables do not have multivariate normality or when they have excessive kurtosis, it is desirable to employ the WLS estimation method. In order to determine an appropriate estimation method, the normality of the variables should be statistically tested.

The fundamental concept in estimating the SEM is that the population covariance matrix of observed variables (Σ) can be expressed in terms of unknown parameter θ, which includes all the unknown parameters in B, Γ, Φ and Ψ matrices. Each element of the population covariance matrix can be written as a function of one or more model parameters, or $\Sigma = \Sigma(\theta)$. Hence, the parameters θ can be estimated by minimizing the discrepancies between the sample covariance matrix S and the population covariance matrix expressed in terms of unknown parameters $\Sigma(\theta)$.

Table 2
Elements of ML and WLS

Σ	The population covariance matrix of observed variables
$\Sigma(\theta)$	The implied covariance matrix of structural parameters
S	The sample covariance matrix
s	A vector of $\frac{1}{2}(p+q)(p+q+1)$ elements obtained by placing the nonduplicated elements of S
$\sigma(\theta)$	The corresponding same-order vector of $\Sigma(\theta)$
W^{-1}	$\frac{1}{2}(p+q)(p+q+1) \times \frac{1}{2}(p+q)(p+q+1)$ positive-definite weight matrix

The maximum likelihood (ML) approach will estimate θ by minimizing the fit function:

$$F_{ML}(\theta) = \log\left|\sum(\theta)\right| + tr\left[S\sum^{-1}(\theta)\right] - \log|S| - (p+q)$$

This fit function assumes that the observed variables have a multinormal distribution.

The WLS approach will estimate θ by minimizing the fit function:

$$F_{WLS}(\theta) = [s - \sigma(\theta)]' W^{-1}[s - \sigma(\theta)]$$

Values of θ are estimated so as to minimize the weighted sum of squared deviations of s from $\sigma(\theta)$ (Bollen, 1989). WLS method does not assume multivariate normality of variables and does need asymptotic covariance of variables for estimation. The elements of ML and WLS are summarized in Table 2.

4. Conceptual framework of model and variables

4.1. Descriptive statistics of data

The data used in this study are 2880 complete accident records during the year 2005, which are collected by Korean Expressway Corporation. Each accident record has various and rich information such as the accident location (where the accident took place), pavement type, horizontal alignment, vertical alignment, vehicle type, driver's gender, driver's age, road surface condition, the day (weekend or weekday), weather condition, day or nighttime, the number of deaths, the number of injured persons, the number of involved vehicles, and the number of damaged vehicles. After eliminating missing and erroneous data, 2649 accident data are utilized in this search.

In Table 3, mean values of four variables such as the number of deaths, the number of injured, the number of involved vehicles, and the number of damaged vehicles is summarized by 11 variables, which may affect on 'accident size' defined in this study. Comparisons of the mean values give meaningful indications to relationships between accident size and several factors. For example, accidents on 'main road' have significantly higher mean values of accident size related variables rather than those on 'others (ramp, tunnel, toll gate, etc.)'. Similarly, the comparison shows that the accident size increases as horizontal alignment is straight, weather condition is clear, surface condition is dry, time zone is nighttime, and driver is male.

Table 3
Descriptive statistics of accident records

	Frequency	Percentage	Mean			
			Number of deaths	Number of injured	Number of involved	Number of damaged
The accident location						
Main road	2046	77.2	0.09	0.44	1.47	0.55
Others (ramp, etc.)	603	22.8	0.06	0.20	1.24	0.40
Pavement type						
Concrete	1415	53.4	0.08	0.37	1.40	0.51
Asphalt	1234	46.6	0.09	0.41	1.43	0.52
Horizontal alignment						
Straight ($R \geq 500$ m)	2419	91.3	0.09	0.39	1.42	0.52
Curve ($R < 500$ m)	230	8.7	0.05	0.33	1.32	0.48
Vertical alignment						
Level	1454	54.9	0.08	0.36	1.45	0.52
Up-slope (0 to +3%)	469	17.7	0.08	0.45	1.40	0.52
Downslope (−3 to 0%)	581	21.9	0.08	0.37	1.34	0.50
Up-slope (>+3%)	56	2.1	0.14	0.39	1.30	0.49
Downslope (<−3%)	89	3.4	0.15	0.55	1.46	0.60
Weather condition						
Clear	1539	58.1	0.09	0.42	1.47	0.54
Others (rain, fog, etc.)	1110	41.9	0.07	0.34	1.34	0.49
Surface condition						
Dry	1884	71.1	0.09	0.41	1.47	0.54
Wet	765	28.9	0.06	0.32	1.28	0.46
Day or nighttime						
Daytime	1614	60.9	0.07	0.38	1.40	0.50
nighttime	1035	39.1	0.11	0.40	1.44	0.55
The day						
Weekends	888	33.5	0.09	0.38	1.36	0.50
Weekdays	1761	66.5	0.08	0.39	1.44	0.52
Vehicle type						
Auto	1621	61.2	0.08	0.42	1.33	0.50
Others (truck, trailer)	1028	38.8	0.09	0.34	1.55	0.53
Driver's gender						
Male	2305	87.0	0.09	0.39	1.44	0.53
Female	344	13.0	0.04	0.36	1.24	0.45
Driver's age						
20–29	444	16.8	0.07	0.37	1.35	0.52
30–39	919	34.7	0.10	0.34	1.40	0.50
40–49	399	15.1	0.09	0.51	1.49	0.53
50–59	806	30.4	0.07	0.39	1.42	0.52
More than 60	81	3.0	0.15	0.37	1.47	0.52

However, these results do not mean that the poor conditions (e.g., curve or slope) and environments (e.g., nasty weather or wet road surface) decrease accidents. For instance, the accident rate in upward and/or downward slope sections together is 45.1% while total length of slope sections is only about 300 km (about 10% of the total highway length). We can conclude that accident occurs frequently in slope sections than level sections. However, once accident occurred, it is much severer on level terrain roads than on slope roads.

4.2. Initial model specification and data description

An initial model is developed using descriptive data analysis shown in the previous section. Eleven variables of accident records are set to '*X* observed variables', which could be split to several groups having similar characteristics (i.e., exogenous latent variables in SEM). Four variables are set to '*Y* observed variables', which could make endogenous latent variables representing 'accident size' in SEM.

In our initial model, 15 observed variables are used and they are split to four groups: road, environment, driver and accident size group. Road group includes the accident location, pavement type, horizontal and vertical alignment characteristics. Environment group has road surface condition, weekends or weekdays, weather condition and daytime or nighttime. The driver group consists of vehicle type, driver's gender and their age. Accident size group is explained by the number of deaths, the number of injured persons, the number of vehicles involved and the number of damaged vehicles. Table 4 shows descriptions and input codes of observed variables.

All categorical and nominal variables are transferred to binary variables because it is a proper way to deal with categorical and nominal variables in SEM and they allow us to identify the nonlinear influence of categorical and nominal variables on endogenous variables in SEM. Initially, we create all possible binary variables for each categorical and nominal variable and test statistical significance of variables.

For driver's age, five binary variables are created and tested to construct exogenous measurement model: (1) 20–29: 1, others: 0; (2) 30–39: 1, others: 0; (3) 40–49: 1, others: 0; (4) 50–59:1, others: 0; (5) more than 60: 1, others: 0. Among the five binary variables, only (1) and (3) show the statistical significance so that those two variables are included in final SEM.

Table 4
Definitions of variables and their codes

Group	Observed variables	Description and coding input value
Road group	The accident location	The main road: 1 Others (ramp, tunnel, bridge, toll gate, etc.): 0
	Pavement type	Concrete: 1 Asphalt: 0
	Horizontal alignment	A straight section: 1 A curve section: 0
	Vertical alignment	Up-slope (more than +3%): 1 Others: 0
Environment group	Weather condition	Clear: 1 Others (rain, fog, snow etc.): 0
	Surface condition	Dry: 1 Wet: 0
	Day or night	Nighttime: 1 Daytime: 0
	The day	Weekday (Monday to Friday): 1 Weekend (Saturday and Sunday) or public holiday: 0
Driver group	Vehicle type	Truck, trailer: 1 Auto, van: 0
	Driver's gender	Male: 1 Female: 0
	Driver's age	20–29:1 Others: 0 40–49:1 Others: 0
Accident size group	The number of deaths	(Persons, ordered variable)
	The number of injured persons	(Persons, ordered variable)
	The number of involved vehicles	The number of vehicles that were associated with the accident (units, continuous variable)
	The number of damaged vehicles	The number of vehicles that were damaged (units, continuous variable)

Y observed variables (i.e., size group) are treated as ordered variables or continuous variables. Since the number of involved vehicles and the number of damaged vehicles variables have more than 15 categories, LISREL program treats them as continuous variables. The number of death and the number of injured persons are set to ordered variables. The four groups defined at this stage will be re-classified based on results of factor analysis described in Section 5.

5. Development of final SEM

The final model specification is derived using a two-stage development process. At the first stage, we conduct factor analysis to classify observed variables into several groups. Factor analysis is often used to analyze the correlations among several variables in order to estimate and to describe the number of fundamental dimensions that underlie the observed data. Those fundamental dimensions (factors) can be latent variables in SEM. At the second stage, we estimate the polychoric correlations matrix of observed variables and finally develop a SEM having the best-fit statistic.

5.1. Factor analysis

Factor analysis is performed on 12 *X* observed variables, based on the result of which exogenous latent variables is determined. The results of the factor analysis with un-rotated and orthogonally rotated are shown in Tables 5 and 6, respectively. Table 5 contains the un-rotated factor loadings, which are the correlations between each variable (rows) and each factor (columns). Loadings above 0.6 are usually considered 'high' and those below 0.4 are 'low.' For example, the first factor can be called 'environment factor' because items like weather condition and road surface condition load highly on it. The second factor can be called 'driver factor' because vehicle type and driver's gender have high loadings for the factor. The third factor called 'road factor' is associated with horizontal alignment and vertical alignment variables. Other observed variables are classified into fourth, fifth and sixth factors. Generally, Varimax rotation is a common technique, which attempts to minimize

Table 5
Un-rotated factor analysis results

	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6
Accident location	−0.112	−0.338	0.605	−0.057	0.222	0.087
Pavement type	0.097	0.144	0.396	0.000	−0.440	0.457
Horizontal alignment	0.082	0.006	0.705	0.126	−0.193	−0.057
Vertical alignment	−0.044	0.030	−0.138	−0.026	0.369	0.858
Weather condition	0.867	0.179	−0.015	0.029	0.035	0.009
Road surface condition	0.893	0.130	−0.005	−0.030	0.040	0.010
Daytime or nighttime	0.118	−0.007	0.289	0.129	0.743	−0.200
The day (weekdays or weekends)	−0.108	0.312	0.114	0.204	−0.130	−0.060
Vehicle type	−0.052	0.695	−0.024	0.411	−0.011	−0.005
Driver's gender	−0.245	0.380	0.032	0.595	0.159	0.031
Driver's age in 20s	0.173	−0.597	−0.103	0.489	−0.106	0.058
Driver's age in 40s	−0.116	0.481	0.123	−0.630	0.097	−0.043

Table 6
Varimax rotated factor analysis results

	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6
Accident location	−0.194	−0.242	0.575	−0.045	0.338	0.094
Pavement type	0.123	0.112	0.541	0.053	−0.463	0.229
Horizontal alignment	0.062	0.116	0.711	−0.016	0.027	−0.193
Vertical alignment	−0.010	−0.021	−0.056	0.012	0.017	0.944
Weather condition	0.886	0.000	0.001	−0.030	0.031	−0.003
Road surface condition	0.900	−0.079	0.014	−0.019	0.035	−0.003
Daytime or nighttime	0.113	0.057	0.101	0.029	0.822	0.050
The day (weekdays or weekends)	−0.039	0.391	0.097	0.067	−0.079	−0.098
Vehicle type	0.109	0.786	−0.075	0.139	−0.032	0.021
Driver's gender	−0.147	0.708	−0.017	−0.142	0.175	0.104
Driver's age in 20s	0.036	−0.142	0.024	−0.793	−0.010	−0.004
Driver's age in 40s	−0.008	−0.044	0.006	0.816	−0.004	0.007

Factor 1 (environment factor): weather condition, road surface condition, daytime or nighttime. Factor 2 (driver factor): vehicle type, driver's gender, driver's age in twenties, driver's age in forties. Factor 3 (road factor): accident location, pavement type, horizontal alignment, and vertical alignment. Factor 4, 5 and 6 (the others): the day (weekdays or weekends).

Table 7
Test of univariate normality for continuous variables

	Skewness		Kurtosis		Skewness and kurtosis	
	Z-score	*P*-value	*Z*-score	*P*-value	Chi-square	*P*-value
# of involved vehicles	58.484	0.000	164.422	0.000	30454.854	0.000
# of damaged vehicles	64.789	0.000	144.511	0.000	25080.968	0.000

Relative multivariate kurtosis = 4.832.

the complexity of the factors by making the large loadings larger and the small loadings smaller within each factor. Varimax rotated matrix is shown in Table 6. Pavement type loaded highly on sixth factor in un-rotated analysis is re-classified into third factor by rotated component factor analysis.

Several variables of factor 4, 5 and 6, which are regarded as significant exogenous observed variables are re-classified into factors 1–3. For example, driver's ages in twenties and forties were classified into the factor 4 through factor analysis. They were, however, re-classified into 'driver factor (factor 2)' because they are driver's characteristics. Vertical alignment and daytime or nighttime are similar to this case, and we re-classify that into 'road factor' and 'Environment factor.' The day (weekdays or weekends) variable was also re-classified into 'the other factors (factor 4–6)' because this variable does not fit in 'driver factors (factor 2)' and they have small loadings (less than 0.4).

Finally, three factors are used with exogenous latent variables in the model. Fourteen observed variables (10 *X* observed variables and four *Y* observed variables) into four latent variables (three exogenous and one endogenous variables) for SEM are classified based on the result of factor analysis. Exogenous latent variables are factor 1 (environment factor), factor 2 (driver factor) and factor 3 (road factor). Endogenous latent variable is "accident size factor".

5.2. A SEM for accident size in Korea highway

Total 2649 accident samples are analyzed and correlation matrix among 14 variables is created using PRELIS software considering the variable distribution characteristics. A fundamental principle in PRELIS is the distinction between variables of different scale types. All *X* observed variables are binary (two ordinal) variables and *Y* observed variables are continuous or ordinal variables in our model so that we use polychoric correlations matrix computed by PRELIS. PRELIS also produces estimates of the asymptotic covariance matrix. This can be used to compute a weight matrix (*W*) for WLS in LISREL (Joreskog and Sorbom, 2000).

Table 8
Test of multivariate normality for continuous variables

Skewness	
Value	17.664
Z-score	48.994
P-value	0.000
Kurtosis	
Value	38.659
Z-score	33.775
P-value	0.000
Skewness and kurtosis	
Chi-square	3541.208
P-value	0.000

Univariate and multivariate normality is tested to determine the estimating method. PRELIS gives univariate and multivariate tests of normality for continuous variables and the results are showed in Tables 7 and 8, respectively. We know that the standard normal variate is our *Z*-score. 95% area under the curve lies between −1.96 and +1.96, 99% lies between −2.58 and +2.58. Therefore, we can reject the hypothesis that the distribution has multivariate normality because *Z*-scores of skewness and kurtosis exceed

Table 9
New coding values of variables

Variable	Coding input value	Variable	Coding input value
Pavement type	Concrete: 1, asphalt: 0	Day or nighttime	Night: 1, day: 0
Horizontal alignment	Straight: 1, curve: 0	Vehicle type	Truck: 1, auto: 0
Vertical alignment	Up-slope (>+3%): 1, others: 0	Driver's gender	Male: 1, female: 0
Weather	Clear: 1, others: 0	Driver's age (20s)	20s: 1, others: 0
Road surface	Dry: 1, wet: 0	Driver's age (40s)	40s: 1, others: 0

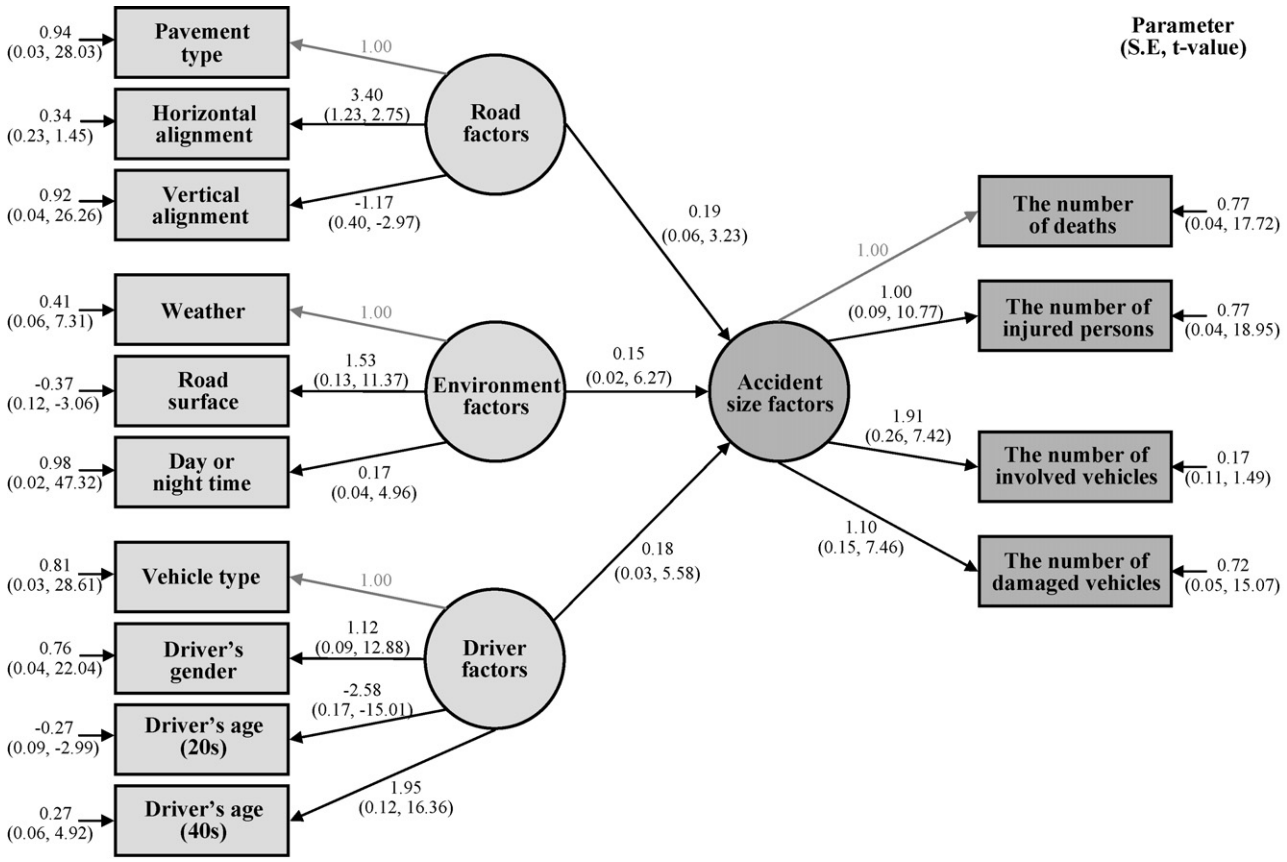


Fig. 2. Final structural equation model of traffic accident size.

Table 10
95% Confidence interval of lambda-*X*

Observed variables	Road factor	Driver factor	Environment factor
Pavement type	(Reference variable)		
Horizontal alignment	$0.98 \le \lambda \le 5.81$		
Vertical alignment	$-1.95 \le \lambda \le -0.39$		
Weather condition		(Reference variable)	
Road surface condition		$1.28 \le \lambda \le 1.78$	
Daytime or nighttime		$0.09 \le \lambda \le 0.25$	
Vehicle type			(Reference variable)
Driver's gender			$0.94 \le \lambda \le 1.30$
Driver's age in twenties			$-2.91 \le \lambda \le -2.25$
Driver's age in fifties			$1.71 \le \lambda \le 2.19$

these values. 'In general, ML (maximum likelihood) estimation is fairly robust against violations of multivariate normality for sample sizes commonly encountered in transportation research (Golob, 2003).' WLS does not assume multivariate normality and is known as the asymptotically distribution free. For our cases, WLS estimation method is employed because distributions of variables do not have multivariate normality as shown in Tables 7 and 8.

The final SEM of traffic accident size is depicted in Fig. 2. For easier interpretation of the model, new values of each variable are recoded as shown in Table 9. As shown in Fig. 2, road factors, driver factors and environment factors are selected as exogenous latent variables and accident size factors are set as endogenous latent variables. In the figure, the numbers in the arrows are parameters estimated and numbers in parentheses indicate standard error and *t*-value, respectively. In Tables 10–12, there are 95% confidence intervals for the coefficients estimated. Latent variables are unobservable and do not have definite scales so that the unit of measurement in each latent variable is arbitrary. To define the

Table 11
95% Confidence interval of lambda-*Y*

Observed variables	Size factor
The number of vehicles involved	(Reference variable)
The number of damage vehicles	$0.82 \le \lambda \le 1.18$
The number of deaths	$1.36 \le \lambda \le 2.46$
The number of injured persons	$0.81 \le \lambda \le 1.39$

Table 12
95% Confidence interval of gamma

Structural model	Size factor
Road factor	$0.17 \le \lambda \le 0.31$
Driver factor	$0.11 \le \lambda \le 0.19$
Environment factor	$0.12 \le \lambda \le 0.24$

Table 13
Total effect of KSI (exogenous variables) on ETA (endogenous variables)

	Road factor (S.E., *t*-value)	Environment factor (S.E., *t*-value)	Driver factor (S.E., *t*-value)
The number of deaths	0.19 (0.06, 3.23)	0.15 (0.02, 6.27)	0.18 (0.03, 5.58)
The number of injured persons	0.19 (0.06, 3.21)	0.15 (0.02, 6.31)	0.18 (0.03, 5.66)
The number of involved vehicles	0.37 (0.11, 3.50)	0.29 (0.03, 9.32)	0.34 (0.05, 7.56)
The number of damaged vehicles	0.21 (0.06, 3.61)	0.17 (0.01, 12.91)	0.20 (0.02, 8.89)

Table 14
Goodness of fit statistics

Fit index		Fit index	
Chi-square	959.719 (P = 0.000)	AGFI (adjusted goodness of fit index)	0.997 (0.9 and more)
RMSEA (root means square error of approximation)	0.069 (0.05 and less)	CFI (comparative fit index)	0.998 (0.9 and more)
RMR (root mean square residual)	0.113 (0.05 and less)	CN (critical N)	281.400 (200 and more)

unit of measurement of each latent variable 'reference variables' is adopted. A non-zero coefficient (usually one) is given to one of observed variables as an indicator (i.e., reference variable). This reference variable can give same unit of observed variables to latent variables. Hence, reference variables do not have *t*-values because they are not estimated.

The estimated coefficients are all standardized solutions, so we can compare the effect of each variable on latent variables. In *X* measurement model, for example, road surface condition (1.53) has a stronger influence on the environment factor than weather condition (1.00) and day or nighttime (0.17) variable. Horizontal alignment variable has about three times stronger effects (3.40/1.00) on road latent variables compared with pavement type. In structure model, parameters estimated of three exogenous latent variables shows that the main factors influencing on the accident size are the road factors even though the difference of effects is relatively small. Hence, in order to decrease the traffic accident size handling the road factor is more effective than handling driver and environment factors. It can be a positive result to traffic engineers because as they can handle 'road factors', they hardly manage 'driver and environment factors'. In *Y* measurement model, 'accident size factor' is mostly described by the number of vehicles involved.

The total effect shown in Table 13 is the sum of indirect effect and direct effect. Since our model does not have indirect structures the total effect is equal to direct effect in SEM.

The final model has 14 observed variables, three exogenous and one endogenous latent variables involving interactions between them. Through the final structure equation model we find the relationship between the accident size and each observed variable. Various factors determine accident size, furthermore, all *X* observed variables and *Y* observed variables have significant effects (high *t*-values) on latent variables.

The model tells us that up-slope section (more than +3%) and curve section can help to decrease the number of death, the number of injured, the number of involved vehicles and the number of damaged vehicles. The results should be highly related to operating speeds on the highway. Several studies for operating speeds on the highways can be found in Korea and other countries. Fitzpatrick and Collins (1999) shows that operating speeds on up-slope (4–9%) and curve sections (R is smaller than 500 m) is much slower than level terrain and straight sections. Similar studies have been conducted in Korea. Lee et al. (2006) have collected operating speed data in Korean Highway. The study showed that the average operating speed in straight sections was 108.89 km/h (N = 528, S = 72.21) while the average operating speed in curve sections was 103.44 km/h (N = 320, S = 102.35). They also concluded that up-slope vertical alignment (greater than 2%) leads to the decrease operating speed using a regression model.

In addition, SEM results show that clear weather, dry road surface lead to increase accident size. We think that the higher operating speed due to driver's carelessness on a clear day and slower operating speed in bad weather can explain the result. This result has a consistency with other researches. For instance, Choi and Son (1999) analyzed the effect of rain on traffic flows in urban freeway basic segments. The average service flow rate (it can be interpreted as a capacity of freeway) was reduced in rainy condition and the operating speed also was reduced about 10–20%. In case of Driver's factor, drivers in forties (compared to other ages), male (compared to female) and truck vehicles (compare to autos) leads to increase accident size. We can conclude that various factors affect on the operating speed and operating speed is a major factor to determine the accident size.

The estimated model possesses 71 degrees of freedom, with a Chi-square value of 959.719 (P = 0.000). It is known that Chi-square value is so sensitive of sample size that *P*-value has low value along with increasing sample size (more than 200). The sample size of our model is so large (2,649) that Chi-square value is high. In SEM approach, therefore, the goodness of fit is generally performed by using other criteria such as RMSEA, RMR, AGFI, CFI and NFI. It is generally accepted that the value of RMSEA (root mean square error of approximation, $\sqrt{}$population discrepancy function value/degree of freedom) for a good model should be less than 0.05, but there are strong arguments that the entire 90% confidence interval for RMSEA should be less than 0.05. The root mean square residual (RMR, $\sqrt{(1/k)\Sigma_{ij}(s-\sigma)^2}$) is index based on the direct comparison of the sample and model implied variance–covariance matrix include. RMR with values less than 0.05 being considered a good fit. GFI and AGFI do not depend on sample size explicitly and measure how much better the model fits as compared to no model at all. Both of these measures should be between zero and one, the recommended acceptable value is 0.9 and more. CFI and NFI measures how much better the model fits as compared to a baseline model, and these indices are supposed to lie between 0 and 1. The goodness of fit statistics for our model are presented in Table 14.

6. Conclusions

In this research, we postulated that road factors, driver factors and environment factors are exogenous latent variables and accident size factor is an endogenous latent variable for SEM to analyze traffic accidents size. The observed variables for latent variables are pavement type, horizontal and vertical alignment characteristics, weather condition, road surface condition, daytime or nighttime, vehicle type, driver's gender and their age and forth. Using factor analysis, the 14 variables are grouped into four latent variables (three exogenous and one endogenous variables) for SEM.

The SEM illustrates positive or negative effects of each variable on the accident size. According to the SEM model, the total effect of road factors on accident size is 0.19 (S.E. 0.06, *t*-value 3.23), so that accident size tends to increase when road factors have higher values. Road factors increase in case of pavement of concrete, straight and level/downward slope. The results are consistent with results of previous researches since the operating speed is slower in poor (curve or upward slope) section of roads. The estimated coefficient of environment factors is a positive value (0.15, S.E. 0.02, *t*-value 6.27). This result indicates that poor weather and wet road surface contribute to decrease accident size. In case of driver factors, the estimated coefficient is 0.18 (S.E. 0.03, *t*-value 5.58), which means that auto vehicle and female drivers contribute to decrease accident size.

The estimated coefficients are all standardized solutions, so we can conclude that the major factors influencing on the accident size is road factors. As previously stated, these results do not mean that the poor sections (curve or slope) and environments (nasty weather or nighttime) decrease accidents. Said in another way, obviously under poor designed sections and environment conditions, accident occurs more frequently. However, once accident occurred in straight sections, the accident size is higher than curve or up-slope ones. Factors increasing the accident size are related to operating speed and driver's carelessness.

Among three exogenous latent variables (road, environment and driver factors), the effect of road factor on accident size is highest. In order to decrease the traffic accident size handling the road factor is more effective than handling driver and environment factors. It can be a positive result to traffic engineers because as they can handle 'road factors', they hardly manage 'driver and environment factors'. The findings in this research offer information about the relationships between accident size and various factors and they can contribute to reduce traffic accident size. Although there are countless factors having relation to "accident size", obtainable information from fields is very limited. Hence, while some aspects are not properly described and explained by models, we still find valuable information from models developed in this study. In future studies, new latent and observed variables such as enforcement data, information on VMS and signs, and so forth affecting traffic accidents should be included in the model.

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